

Differential Calculus

*Home work sheet # 5

Successive Differentiation

(a) Find the n -th derivative of the following functions:

1. $y = x^n$

2. $y = (ax + b)^n$

3. $y = \ln(ax + b)$

4. $y = \frac{1}{x + a}$

5. $y = e^{ax}$

6. $y = \sin(ax + b)$

7. $y = \cos(ax + b)$

(b). If $y = e^{ax} \sin bx$, then show that $y_2 - 2ay_1 + (a^2 + b^2)y = 0$.

(c). If $y = e^x \sin x$, then show that $y_4 + 4y = 0$.

Rolle's and Mean Value Theorem

(a). Verify the hypothesis of Rolle's Theorem for the following functions:

1. $f(x) = x^2 - 6x + 8$; $[2, 4]$

2. $f(x) = \cos x$; $[\pi/2, 3\pi/2]$

3. $f(x) = \frac{x}{2} - \sqrt{x}$; $[0, 4]$.

(b). Verify the hypothesis of Mean Value Theorem for the following functions:

1. $f(x) = x^3 + x - 4$; $[-1, 2]$

2. $f(x) = \sqrt{x+1}$; $[0, 3]$

3. $f(x) = \sqrt{25 - x^2}$; $[0, 5]$.

Maclaurin and Taylor Series

(a). Find the Taylor series for the following functions:

(i) $\sin x$, at $x_0 = \frac{\pi}{2}$. (ii) $\ln x$, at $x_0 = 2$.

(b). Expand $y = \ln x$ in the power of $x - 2$ and $y = e^{ax}$ in the power of $x - 1$.

(c). Find the Maclaurin series for the function e^{ax} and $\cos x$.

(d). Find the Maclaurin polynomial p_0, p_1, p_2, p_3 for $e^x \cos x$.

(e). Expand $y = \ln(x + 1)$, $y = \sin x$, $y = \cos x$ in the power of x .

**These problems are for the students only as home work. Search the reference books for more examples.*